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Master Thesis in Natural Computation

# Bio-inspired learning for motion planning of collaborative robots

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Then what is your hand for?

---

*Rei Ayanami, End of Evangelion*

To effectively contain a civilization's development and disarm it across such a long span of time, there is only one way: kill its science.

---

*Liu Cixin, The Three-Body Problem*

If one's different, one's bound to be lonely.

---

*Aldous Huxley, Brave New World*

The saddest aspect of life right now is that science gathers knowledge faster than society gathers wisdom.

---

*Isaac Asimov, Book of Science and Nature Questions*

**אמת**

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*The Golem of Prague*

The true measure of a man is not his intelligence or how high he rises in this freak establishment. No, the true measure of a man is this: how quickly can he respond to the needs of others and how much of himself he can give.

---

*Philip K. Dick, Our Friends*

# TODO

Restructure the document, according to the following outline:

- How many paths shall be shown and executed?

  What are they?

- How many repetitions shall be performed for each path?
- What shall the participant be told when entering the experimental room? Provide written protocol.
- What information shall be encoded in a waypoint?
- How can waypoint information be effectively represented in 3D environment? What are the graphical markers aiding the participant?
- How many trials shall be performed for the same path prior to the data collection session?
- What execution information shall be collected?

  What will be regarded as input to task execution?

  What will be regarded as output?

- What kind of processing shall the data undergo before they can be used for NN training?
- What is the data collection mechanism, i.e. set of software components that are exercised to collect data?

# Acknowledgements

Write acknowledgements here.

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# Introduction

Motion coordination is amongst the hardest challenges in interactions between humans, requiring various cognitive and motor skills to work together. In human-robot interaction, the challenge is even greater, since the robot must be able to understand and predict the humans' movements, while also adapting its own movements to their dynamics.

Keeping in mind that humans often use other mean of communications to tell their intentions, such as gaze, facial expressions, gestures, words and so on, the robot must be able to overcome the lack of these communication channels, by relying only on the humans' movements.

This interaction is carried indirectly through touch between the humans and the robot, by making use of some object carried by both of them, or directly through touch, by making use of the robot's end-effector. The word "touch" has many definitions, the most concrete and simple one is the physical contact between two bodies, but it carries a deeper meaning, forgotten through the evolution of the human species: the ability to communicate and understand each other through the sense of touch. It allowed humans to survive by forming bonds when language wasn't yet available.

Reshaped and suppressed through the evolution of the societal structures and hierarchies, the sense of touch has been relegated to a secondary role. To worsen the situation, the last pandemic has forced humans to avoid physical contact for a time enough to forget the importance of it, opening a wound we yet have to heal.

This inadequacy of expressing ourselves through touch reflects on the way new humanoid robots are designed, often equipped with a touch screen rather than a touch-sensitive area on their body. While facials details are overengineered to express emotions, the robot's body is often designed to be a rigid and cold object, rather than an active participant in the interaction.

Endowing robots with the ability to move in a human-like way, to touch and be touched, to communicate and understand each other through the sense of touch, is a challenge that requires a deep understanding of human motion and the ability to replicate it in a robot. A predefined interaction protocol is not enough to achieve this goal, since it doesn't allow the robot to adapt to the human's movements, neither to predict them.

This aspect does not only regard collaborative, industrial related tasks, but also social and assistive robotics, where the robot has to understand the human's feeling and emotions, and adapt its behavior on the fly, to provide a more natural and comfortable interaction. A mandatory

capability in hospital and elderly care.

# 1 State of the art

The following section discusses the current advancements in human-robot interaction, and the limitations of the current literature. The goal is to provide a comprehensive overview of the state of the art in this field, highlighting the key challenges and opportunities for future research.

## 1.1 Current research in human-robot interaction

Some studies have already defined a way to quantify the comfort of a robot's motion [1], others have investigated humans ability to detect if a robot movement reflects a human-like motion or not, comparing sight and proprioception skills [2]. Intent communication through motion is a well-studied topic [3] a robot could tell in advance its plan through its movements to become predictable and thus more comfortable to interact with. Some research have already been done in that way, by evaluating the human signature and replicating in a collaborative task with a simulated user [4].

First, to teach a robot to move in a human-like way, it is necessary to understand how humans move. The theory of kinematics of rapid human movements describes motion as a sequence of lognormal strokes, motor primitives generated by the Central Nervous System, treated as building blocks of complex movements that can be superimposed [5]. By understanding the underlying principles of human motion, a software has been developed that extracts the lognormal parameters from human movements [6]. This software can be applied over a dataset of human movements to extract the lognormal parameters and then use them to generate new movements that are similar to the original ones through a generative model, which could be a neural network or a more biologically plausible model.

Then, it could be possible to study how to adapt the robot's motion to the human partner's dynamics, by integrating bio-inspired adaptive mechanisms that modulate the robot's movement in response to human actions. This could reduce both physical and cognitive workload during interaction, making the collaboration more efficient and comfortable for both parties. This information could be given by teaching the model to predict the human's next movement through the leader follower dynamics, and adapt the movement signature to the human's one. If the human signature is not available, the robot could learn it through interaction, by observing their movements and adapting its own motion accordingly.

However, while the theory of kinematics of rapid human movements provides a framework for understanding human motion, the planning of robot motion is a complex problem that requires the integration of multiple factors.

## 1.2 Literature limitations

# 2 Proposed Solution

This chapter gives a comprehensive overview of the solution adopted to address the problem at hand, spanning from the experimental protocol chosen, the data collection system described in its software and hardware components, to the data processing tools. The goal is to provide a clear understanding of the choices, weighing pros and cons, so to justify the decisions made.

## 2.1 Methodology

For this problem, various software and hardware components are required, ranging from the collection of the data to its processing and analysis, then to the generation of new data and its execution on the very same system that collected the original data.

In the data collection mechanism, the system must interface itself with a human subject, showing them a path to follow and collecting their movements as they execute the task. The result of this process is a dataset that contains the various trajectories of the subjects' movements, which can be used to analyze their behavior and extract useful information for the generation of new movements.

Rather than giving the raw data to a neural network, which would slow down the learning process by increasing its complexity, the data is processed to extract more meaningful information, which will speed up the learning process and make it more efficient. So, by the combination of a theoretical model and data-driven techniques, a bio-inspired generative model can be developed.

The generated movements can then be executed on the same system that collected the original data, allowing for a "hand to hand" comparison between the original and generated movements, through the dyadic interaction between the human subject and the system.

## 2.2 Task Design

The goal of the experiments is relatively simple: collect a dataset of rapid and natural human movements. However, in order to obtain a dataset of sort, it is mandatory to provide the same task to each subject. Nonetheless, the same input must be grasped equally by different subjects. Thus, the deletion of ambiguity in task definition is imperative.

### 2.2.1 Waypoint Definition

As previously mentioned, a path is defined as an ordered sequence of points in the space, called waypoints, each defined by its position in the space. Visually, they are represented through spheres, which size is a trade-off between the accuracy of the collected data and the naturalness of the user's movements. However, even though the user should try to follow the path by crossing the waypoints, some deviation from them is to be expected, since no human movement ever perfectly follows a path. In the same way, a generative model should think of waypoints as a reference, akin to an attractor, rather than a strict constraint.

A path of two waypoints is simply a straight line between the two. On the other hand, a path of three waypoints is interpretable as a curve, since the user is supposed to follow the path by trying to reach the waypoints, but not strongly forcing themselves to cross and step on them: the result is a curve that starts from the first waypoint, gets near the second one, and ends almost on the third one. The same principle applies to paths of more than three waypoints, which are interpreted as a sequence of curves, which curvature is inversely related to the alignment of the waypoints.

The issue with such a representation is that it doesn't provide any information about the curvature of the path, neither about the depth of the waypoints in the space.

### 2.2.2 Path Description

A path, in simple terms, is an ordered sequence of waypoints. This description would be sufficient for a model, since its perception of a three-dimensional space is mathematical and thus unmistakable. However, a human's perception of a three-dimensional space is not mathematical, and thus it is not as accurate as the model one. Therefore, it is necessary to provide the human subject with a more intuitive representation of the path.

A path could represent a common task that a human subject may perform in their daily life, such as moving an object from one place to another, or using a particular tool. The path should be designed to be as intuitive and natural as possible, eventually combining the geometrical description with a semantic one, ensuring that the subject grasps and performs it smoothly, avoiding unnatural movements that could poison the collected data.

**Visual & Haptic Feedback** An intuitive representation should make use of the human natural instinct to merge different sensory inputs to form a more complete perception of the environment. In this case the two only classes of sensory inputs available are the visual and haptic ones, since they are the only two that are directly transformed into positional information in the perceived space by the motor cortex.

The visual feedback is shown to the subject through the ordered sequence of spheres projected in a simulated three-dimensional environment in the screen. The depth perception is

given by the perspective of the camera, which can be moved by the subject to change their point of view.

The haptic feedback is given to the subject through the haptic device's actuators, which apply forces to the subject's hand to guide them towards the path. These forces are applied toward the next waypoint, and let the user stay on each waypoint to give them time to memorize the positions of the waypoints. The more the user's ability to follow the path increases, the less the haptic feedback is applied. More importantly, the rest time on the waypoints is zeroed, since the user must follow the path without stopping on the waypoints. This should happen some time before the haptic feedback is removed, in order to let the user get used to the fact that they are supposed to follow the path without stopping on the waypoints.

**Motor Representation** Both the haptic feedback and the visual feedback are not used during the recording phase, but rather in a previous phase called pre-training. During this step, the user is aided in following the path through the visual and haptic feedback, until a certain level of proficiency is reached. This allows the subject to obtain a representation of the path inside their motor cortex, which is a fundamental aspect of human motion. Once this representation is formed, the user starts to follow the path without any aid, in order to test this proficiency, the feedbacks are slowly removed, until they are completely disabled. The user is then asked to follow the path without any aid, while the system collects their movements.

**Curvature Representation** The only issue with representing a path only by a sequence of points is that it only takes into account direct links between waypoints. Humans rapid movements don't go straight from one point to the following one. Instead, they try to store momentum by not stopping on each point, but making curves during the execution. This aspect must be extracted from the dataset and learned by the generative model, since it is a fundamental aspect of human motion.

An idea to make this aspect stand out from the collected data could be the introduction of arrows in the visual environment, although this means crowding it with more information that could be distracting or hard to interpret for the user. Also, this requires defining other tunable parameters, since the arrows would have to be placed at a specific distance from the waypoint it starts from, with a specific length, at a specific size and orientation. However, by only specifying the waypoints, the user is free to interpret the path in their own way, and thus they can execute it in a more natural way, which is what the experiments aim to achieve. In the end, it is mandatory to explain to the user that they are supposed to follow the path by trying to reach the waypoints, but not strongly forcing themselves to cross and step on them, since this would lead to unnatural movements: the user should be encouraged to follow the path smoothly.

### 2.2.3 Path Taxonomy

The paths that will be shown to the subject during the experiment are grouped into the following categories:

- Burst: impulsive movements. Simple paths that help the subject getting used to the haptic device.
  - Straight line over the $XY, YZ$ or $XZ$ planes
  - Semicircumference over the $XY, YZ$ or $XZ$ planes
- Geometric: two- and three-dimensional round geometric shapes
  - Helix, growing over the $X$ or $Y$ axes
  - Spiral, in the $XY$ plane
  - Ellipses, one in the $XY$ plane and the other in the $XZ$ plane
  - Circle, one in the $XY$ plane and the other in the $XZ$ plane
- Symbol: letters of an alphabet
  - Cursive lowercase L $(\ell)$ in both the $XY$ and $XZ$ planes
  - Print uppercase S in an oblique plane in both senses
  - Cursive Greek Phi $(\varphi)$ in the $XY$ plane
  - Print uppercase $N$ which segments differ in plane

## 2.3 Session Design

The real challenge is to design the experiments in such a way that there isn't difference between the information received by the subjects during the experiment and the one received by the generative model during its training phase.

### 2.3.1 Sources of Bias

To ensure consistency across different subjects, the experimental protocol should be designed to minimize any potential biases or confounding factors that could affect the results. For instance, it is necessary to simplify some complex terms by finding more intuitive and user-friendly ways to explain them, so to ensure user comprehension and engagement.

Additionally, the protocol should be designed to minimize any potential sources of error or variability in the data collection process, such as differences in the user's physical abilities or

environmental factors that could affect their movements. This last point is particularly important, since the experiment is designed to be executed in a laboratory environment, where the user is asked to sit on a chair and interact with the system through a haptic device. The laboratory environment should minimize any potential distractions or sources of noise that could affect the user's movements and the data collection process.

**Posture** The environment must be designed to ensure the user's safety and comfort, by providing them with a comfortable chair and ensuring that the haptic device is properly calibrated and adjusted to their physical characteristics (i.e. dominant hand, arm length, height, etc.) to reduce any potential sources of discomfort or fatigue and increase comfort during the experiment. The more the user feels relaxed, the more natural their movements will be, and thus the more accurate the collected data will be.

**Observer effect** The user is aware of the fact that they are being observed and that their movements are being recorded, which could affect their behavior and introduce some bias in the collected data. To prevent this kind of bias, the user should be informed about the purpose of the experiment and the data collection process, and they should be given the opportunity to ask questions and provide feedback. This will help to ensure that the user feels comfortable and engaged in the experiment, and that they are motivated to perform their best. Of course, any reveal of the purpose of the experiment should be done in a way that doesn't influence the user's behavior, and thus bias the collected data.

### 2.3.2 Tunable parameters

The real challenge is to design the experiments in such a way that there isn't difference between the information received by the subjects during the experiment and the one received by the generative model during its training phase.

**Parameters** The session does have various parameters that need to be tuned in a trade-off between the accuracy of the collected data, their naturalness, the fatigue of the user, the length of the experiment and the amount of data collected. The parameters that can be tuned are:

- $W_s$: The size of the spheres that represent the waypoints of the paths, which can be increased to make it easier for the user to follow the path, but at the cost of reducing the accuracy of the collected data. A bigger sphere would require less training and less precision from the user, but it would also make the collected data less accurate, since the user would be able to deviate more from the path without being penalized. On the other hand, a smaller sphere would require more training and more precision from the user, but it would also make the collected data more accurate, since the user would be forced to

follow the path more closely. Although, this forceful guidance could lead to unnatural movements, unnatural to a human, which would be detrimental to the generative model.

- $P_l$: The number of waypoints of a path, which can be increased to make the path more complex, at the cost of increasing the difficulty for the user to follow the path. A more complex path would require more training and heavier cognitive effort, while increasing the data collected. Conversely, a simpler path would require less training and less cognitive effort, while reducing the data collected. The complexity of the path needs to be balanced with the user's ability to follow it, to collect accurate and natural data.

- $P_r$: The number of repetitions for each path, which can be increased to collect more data, at the cost of increasing the fatigue of the user and the length of the experiment. This parameter comes at the cost of making brief rest periods between repetitions, which would compensate the user's fatigue, but at the cost of greatly increasing the time duration of the full experiment session.

- $T_r$: The number of execution prior to the recording phase, named "pre-training phase". Albeit it would be simpler to have a fixed number of repetitions, the issue is within the variability of human learning. Some subjects may learn a path in less time than other, thanks to a prior knowledge of environments over the screen, or of their . Since this parameter depends on the user's ability to follow the path, and on the complexity of the path itself, it is left to decide up to the experimenter. If the user correctly follows the path for a certain number of repetitions, and feels enough confident to proceed to the recording phase, they can switch to the recording phase. Until then, the pre-training will aid the participant by providing the haptic feedback.

- $N_P$: The number of paths, which can be increased to collect more diverse data, at the cost of increasing the length of the experiment. More paths would provide a better representation of the task space and allow to more generalization, but would also require more time and effort from the user. Although, having a few paths would limit the diversity of the data collected, and thus the generalization of the generative model.

- $R_t$ and $R_f$: The rest periods length and frequency (expressed as a fraction of $N_P$), which can be increased to reduce the fatigue of the user, at the cost of increasing the duration of the experiment. The rest periods need to be short to keep the user focused on the task, but long enough to allow them to recover from their fatigue, while not being felt coercive by the user. Moreover, the rest periods need to be frequent enough to allow the user to recover from their fatigue while executing the different paths, but not too frequent to make the user lose focus on the task.

### 2.3.3 Written Protocol

Providing the user with a written protocol is mandatory not only to ensure that all the user have the same information, but also to ensure that they have the same understanding of the task. From the different equivalent sequences of words available to explain an instruction, the most intuitive should be chosen, to avoid any bias in the collected data.

**Input** In each experiment of a session a path is shown to the user, their goal is to traverse close to the waypoint as fast as possible, with the understanding that it's not strictly necessary to cross all the waypoints, but rather to think of them as attractors. They will first be pre-trained to follow the path with the aid of visual and haptic feedback, until they reach a certain level of proficiency, defined by the experimenter, and then they will be asked to follow the path without any aid, while the system collects their movements.

**Result** For each execution of a path, the system collects a sequence of timestamped points in the space, which are expressed in Cartesian coordinates with respect to the reference frame. Furthermore, the tangential velocity of the user's hand is computed at each timestamp. By interpolating these points, and by having a high sampling frequency, it is possible to reconstruct the trajectory of the user's movements with better accuracy. Other information like the user's hand orientation and the angular twist velocity are collected as well, but they are not needed for the generative model, since they are not used to generate new movements. They are collected to provide a more complete representation of the user's movements, and to eventually allow for a more detailed analysis of the data.

```
... Welcome ... What's your dominant hand? (If no preference is
expressed, pick left hand when possible). (Prepare the environment on the
subject choice).

The device to your (left/right) side is a haptic device, which is connected
to the screen in front of you (the screen shows an empty environment). In
the screen, you can observe the effects that manipulating the device has on
the environment.

Please extract the device from its housing and handle it as if it was a
pen or a precision tool. Feel free to explore the environment in the
simulation. Try to exit from it: you will feel forces that will prevent
you from leaving it. Those are the limits within you can freely move.
There are two buttons on the pen: the darker one lets you rotate the
environment, the lighter one to zoom. [mouse]

(Once the subject learnt how to use it). Now, the screen will show you some
colored spheres (inside other transparent spheres?). Try to make a smooth,
fluid and natural movement that crosses all the spheres. The order in which
the movement must cross the spheres is given by their number.

Initially, when you try to cross the spheres, the device will help you reach
them by pulling you towards their center. After a while, the device will
let you traverse to the next sphere, with a slight pulling force.

After a fixed amount of repetitions, the screen won't show any information,
and only the haptic feedback will be left, which pulling force will decrease
execution by execution.

After a fixed amount of repetitions, the pulling force will decrease to zero
(it must dynamically adjust itself by studying the subject accuracy), from
there you will be in the recording phase.

(Should the user be told when they're in the recording phase?)

Every time you want to be placed to the start position and/or reset the
execution, just press the dark button. Now please, try to cross the spheres
with the aid of the device.

[steps: (sight + haptic feedback), (haptic feedback), (no info)]

(When the pre training phase ends) Congrats. Now, proceed to repeat the
same movement you've done before, without the screen information (or
with only the points visible?) as accurately as possible. If you feel
you're off track, don't fix the movement and try again, since it would be
discarded anyways. (In order to help you, the device will automatically
place itself in the starting position when you press the lighter button. A
brief vibration will tell you when to start the movement. If during the
movement you feel the same vibration, then you have to redo it again?)
```

Those are the same inputs and outputs of the generative model, with the exception that it won't follow the same policies, but only understand the underlying principles from the dataset.

## 2.4 System Design

Before delving into the details of the solution adopted, it is necessary to provide a high-level overview of the system design.

### 2.4.1 Haptic Device

The device adopted for the problem is *Touch*, from *3D Systems*. It is a haptic device with six degrees of freedom, allowing it to move in three-dimensional space and rotate around three axes. It is equipped with a stylus that can be held by the user, allowing them to interact with the device. The device is able to produce some linear forces, which are applied to the user's hand through the stylus.

This kind of devices is commonly used in teleoperation, where the user can control a remote robot by moving the stylus and feeling the forces applied by the robot. In this case the device is used to collect data from the user's movements.

### 2.4.2 Haptic Interface Software

The haptic interface software manages both the user interface of the haptic device and the communication with the robot.

From the side of the experimenter, the software component allows them to control the experiment execution, by selecting its configurations, like the path to show, the number of repetitions, the sampling frequency, and so on.

From the side of the human subject, the software component interfaces with them through a simulated environment that projects the position of the haptic device inside a virtual space. In this space, the path is represented as an ordered sequence of spheres, while the device is represented as a sphere that can be moved by the user through their interaction with the device.

#### `ROS2` Controller

The haptic device controller is responsible of managing the communication between experiment, haptic device and subject through sending commands and receiving data, and of the experiment execution.

The controller takes as input the configuration of the experiment, which includes the path to show, the number of repetitions, the size of the waypoints, the forces to apply and so on. From these information, during the execution it is able to communicate with the experiment through the keyboard, and with the haptic device through its command and state interfaces. The controller is also responsible for collecting the data from the haptic device, which includes the position, orientation, linear and angular velocities.

Furthermore, it is responsible of communicating with the subject through the screen via the visualization of the path during the pre-training phase. In this case, the role of the controller is to communicate a series of colored waypoints, and the current position of the haptic device to the visualization software `RViz2`, which is responsible for rendering the scene on the screen. The controller applies the haptic feedback to the user, by computing the forces to apply to the haptic device based on the current position of the device and the position of the next waypoint,

multiplied of a gain factor and the distance between the two.

### 2.4.3 Data Processing Software

The data processing software is responsible for analyzing the collected dataset of human movements and extracting meaningful information that can be used to train a generative model, instead of giving the raw data to the model, which would slow down the learning process by increasing its complexity.

#### Temporal Characterization

In particular, the crucial information that can be extracted from the dataset is the lognormal parameters of each rapid movement across two waypoints, which can be used to generate new movements that are similar to the original ones.

In literature there are already some software components that can extract the lognormal parameters from human movements, such as the *idelog3d* software. Once this software is applied to the dataset, it will output a new dataset that contains the parameters of each rapid movement across two waypoints, which can be learned by the generative model. Once trained, the model can provide the velocity profile of a new movement across two waypoints through the application of the lognormal parameters.

#### Spatial Characterization

Regarding the spatial characterization of the movements, it would be handy to develop a software able to characterize the spatial properties of the movements, such as the starting and ending points, the curvature, the length and so on. In other words, the software should be able to extract the spatial properties of each rapid movement across two waypoints, which can be used to generate new movements that are similar to the original ones.

A possible interpretation of the spatial properties of a movement is to consider it as a Bézier curve, actually a weighted spline Bézier curve, which is defined by a set of control points to which the curve is attracted. The control points are not necessarily the waypoints, while the weights are the parameters that define the attraction of the curve to the control points. The generative model could come with a set of control points(and eventually weights) that define the shape of the curve. In this case, the difference between waypoints and control points is that the former are the points that the model is supposed to reach, while the latter are the points to which the model's curve is attracted to.

For instance, one possible spatial representation is the spline Bézier curve, which comes in handy since it doesn't always reach the target points, rather it tries to get as close as possible to them, which is what humans do when they move. The Bézier curve is defined by a set of control points, which can be used to define the shape of the curve. In order to teach a generative model to draw such lines, it is necessary to extract them from the actual path samples draw from the

subjects movements.

### 2.4.4 Generative Model

The generative model is trained with the dataset of movements, extended with the temporal and spatial characterizations, to learn the underlying principles of human motion. Once trained, the model can generate new movements that are similar to the original ones, by providing the path in the space and its time characterization through a velocity profile derived from the lognormal parameters. The generated movements can then be executed on the same system that collected the original data.

# 3 Experimental Validation

## 3.1 Experimental Setup

Regarding the aforementioned experimental parameters, the following values have been chosen:

| Parameter | Description | Value |
|---|---|---|
| $W_s$ | Waypoint diameter | 0.05 m |
| $P_l$ | Path length | 2-7 waypoints |
| $P_r$ | Path repetitions | 20 repetitions |
| $N_P$ | Number of paths | 20 paths |
| $R_t$ | Rest time | 60s |
| $R_f$ | Rest frequency | $1/6$ |

As explained before, the training repetitions are not a tunable parameter, since they depend on the subject ability to follow the path, and on the complexity of the path itself. At most, it would be possible to define an acceptance criterion during the training phase. Since the training requires them to cross all the waypoints, it is safe to assume that the criterion is in some way already encoded in the waypoint size $W_s$.

## 3.2 Experimental Data

Through the executions of the experiments, the system collects a dataset of human movements, expressed as a sequence of timestamped points in the space, with respect to the reference frame.

## 3.3 Results

## 3.4 Statistical Significance

# 4 Conclusion

Write the conclusion here.

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